Gradient-Boosted Trees and Consensus-Risk

For Opioid and Polypharmacy Emergency Department Visits in Virginia's APCD

R. Jerome Dixon, MDA, MSc IT

Integrative Life Sciences Doctoral Program (Home Department)

School of Pharmacy (Department of Pharmacotherapy & Outcomes Science · Research Lab)

Virginia Commonwealth University

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VCU School of Life Sciences and Sustainability | VCU School of Pharmacy

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Roadmap

QUALIFYING EXAM AGENDA | 5 CORE MODULES

I. Problem

II. Governance & Data

III. Cohorts

IV. Modeling

V. Translation & Deployment

Downstream Clinical & Economic Effects

MODULE I | INFOGRAPHIC PART 1 OF 4 · DOWNSTREAM PUBLIC HEALTH & ECONOMIC TOLL

>500,000 US Opioid Deaths & $78.5B Annual Cost



Since 1999, prescription opioid overdoses and abuse have claimed over half a million lives in the US, generating an economic burden exceeding $78.5 billion annually in healthcare, addiction treatment, and criminal justice expenditures.

Geriatric Polypharmacy: 125,000 Deaths & 700,000 ED Visits Annually

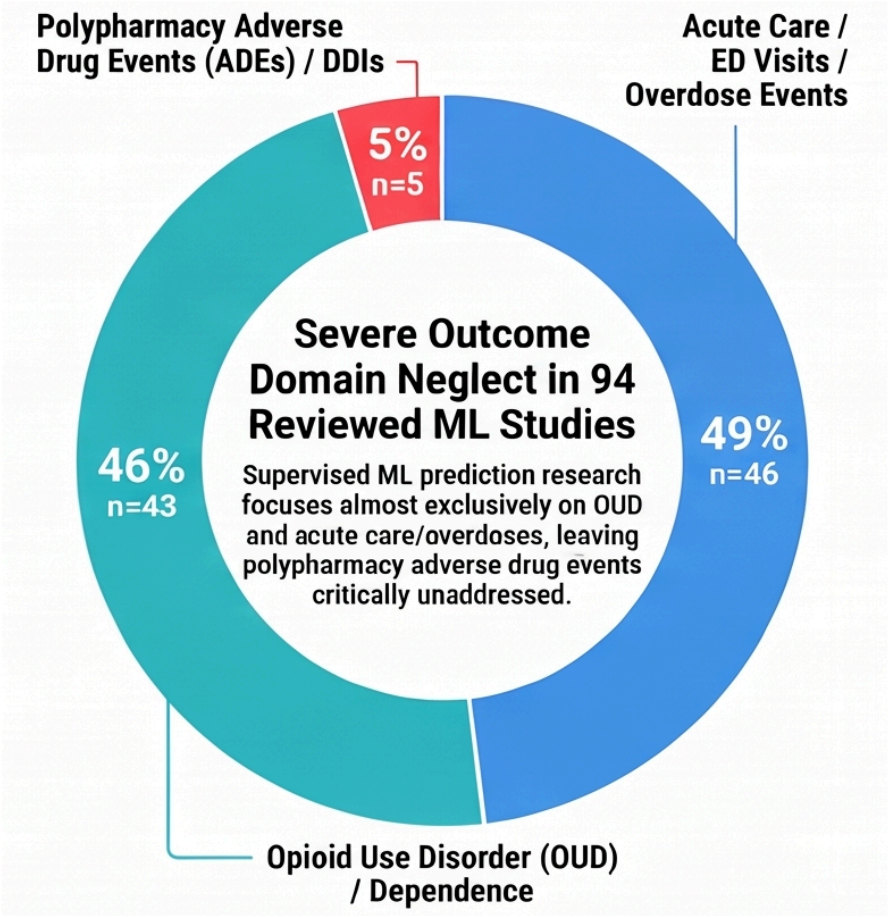


Concurrent use of 5 or more medications affects over 40% of adults aged 65 and older, making polypharmacy the leading driver of preventable adverse drug events in older adults.

Figure 1A (Part 1): Downstream clinical and economic effects of US opioid overdoses and geriatric polypharmacy (CDC WONDER; Florence et al., *Med Care*; Charlesworth et al., *J Gerontol A*; Budnitz/Shehab et al., *JAMA*).

Opioid Research Overshadows Polypharmacy

MODULE I | INFOGRAPHIC PART 2 OF 4 · SYSTEMATIC REVIEW (N = 94 ML STUDIES)





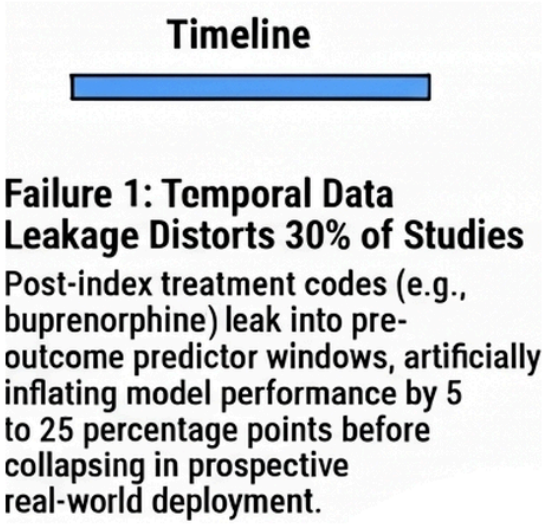
Claims-Based Opioid ED Modeling Void

Out of 1,473 published OUD literature records, only 6 studies specifically evaluate machine learning models designed to predict opioid-related emergency department visits using administrative claims data.

Figure 1A (Part 2): Published outcome domain distribution across 94 supervised ML studies and the claims-based opioid ED modeling void.

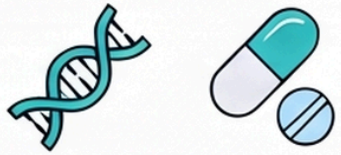
ML Gap – No PGx features. No XAI. No Target Leakage Validation.

MODULE I | INFOGRAPHIC PART 3 OF 4 · PROBAST APPRAISAL (N = 73 MODELS)



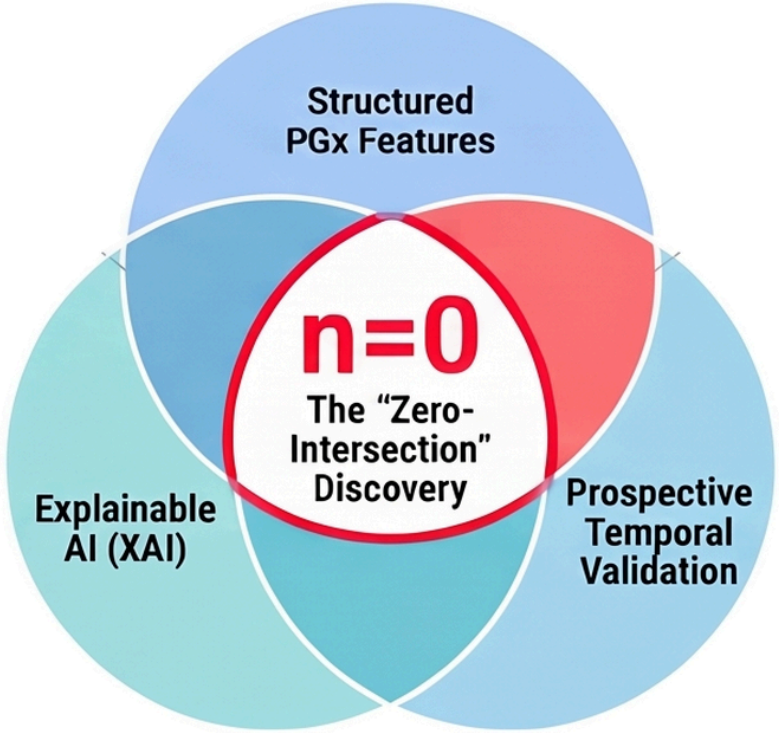
Failure 1: Temporal Data Leakage Distorts 30% of Studies

Post-index treatment codes (e.g., buprenorphine) leak into pre-outcome predictor windows, artificially inflating model performance by 5 to 25 percentage points before collapsing in prospective real-world deployment.



Failure 2: 81% Omit Pharmacogenomics (PGx)

Despite mature CPIC guidelines and known gene-drug modifiers (e.g., CYP2D6, OPRM1), 81% of reviewed ML prediction pipelines completely exclude structured pharmacogenomic features.



Across 73 PROBAST-assessed modeling studies, zero published systems successfully combine structured PGx features, explainable AI (XAI), and prospective temporal validation.

Figure 1A (Part 3): PROBAST appraisal of 73 predictive models exposing the ML gap ($n = 0$ combining PGx features, XAI, and target leakage / temporal holdout validation).

90% of ADEs Occur Within 21 Days

MODULE I | STEP 1

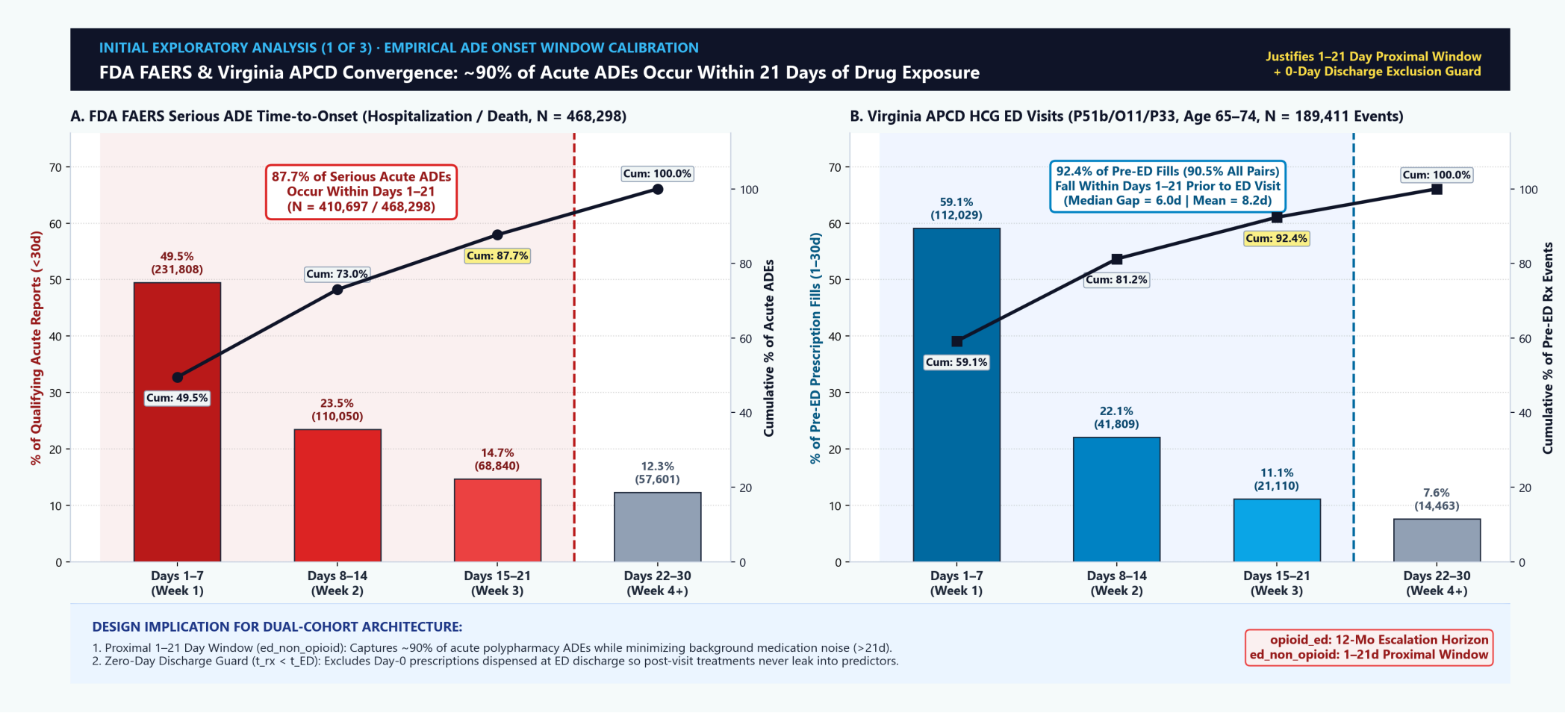


Figure 1B (Part 1): Convergent time-to-onset distribution across FDA FAERS ($N = 468,298$, 87.7% $\leq$ 21d) and Virginia APCD HCG ED visits ($N = 189,411$, 92.4% $\leq$ 21d).

Initial Feature Analysis

MODULE I | STEP 3A

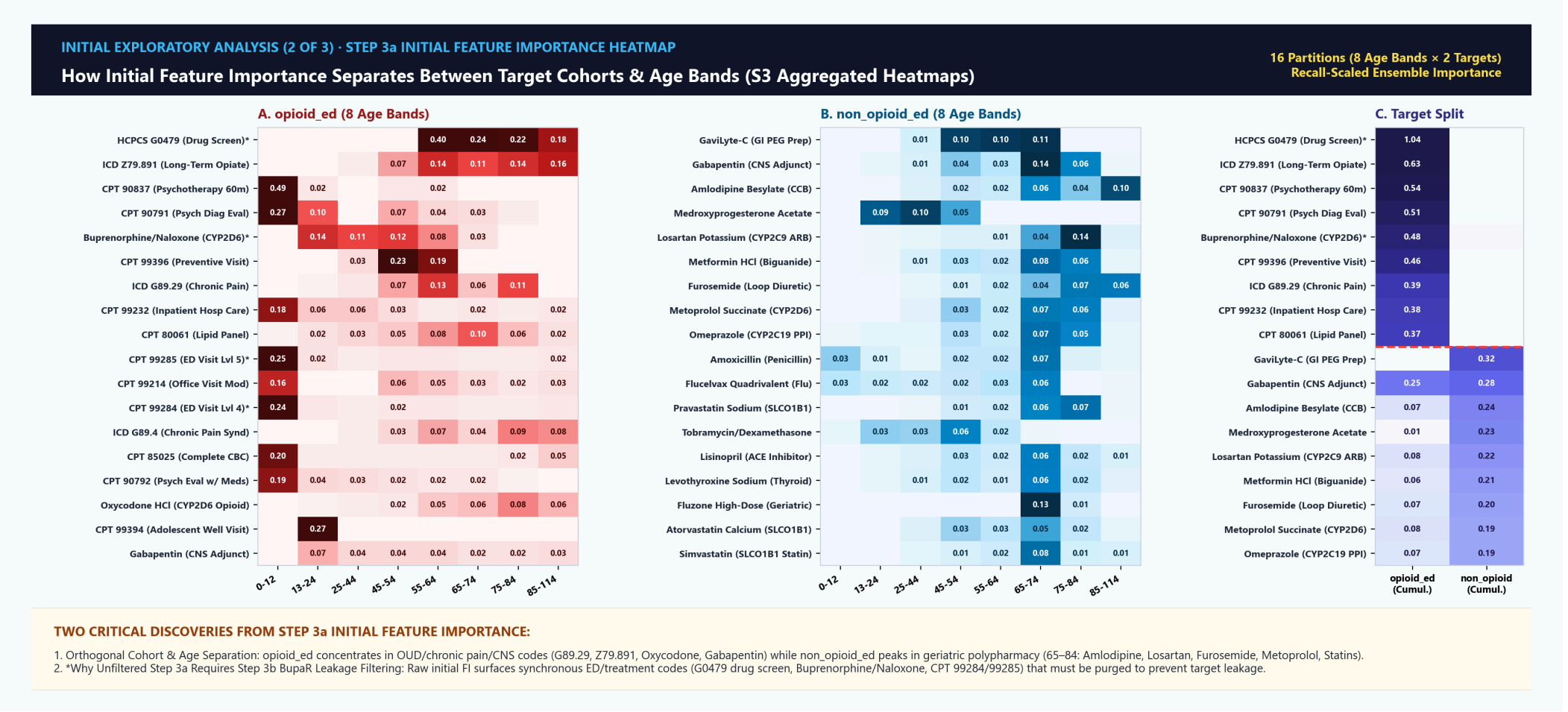


Figure 1B (Part 2): Step 3a multi-model (`CatBoost` , `XGBoost` , `XGBoost-RF`) aggregated feature importance across all 8 age bands (`0-12` to `85-114`) and both targets (`opioid_ed` vs. `non_opioid_ed`).

CPIC Drugs Drive Initial Feature Importance

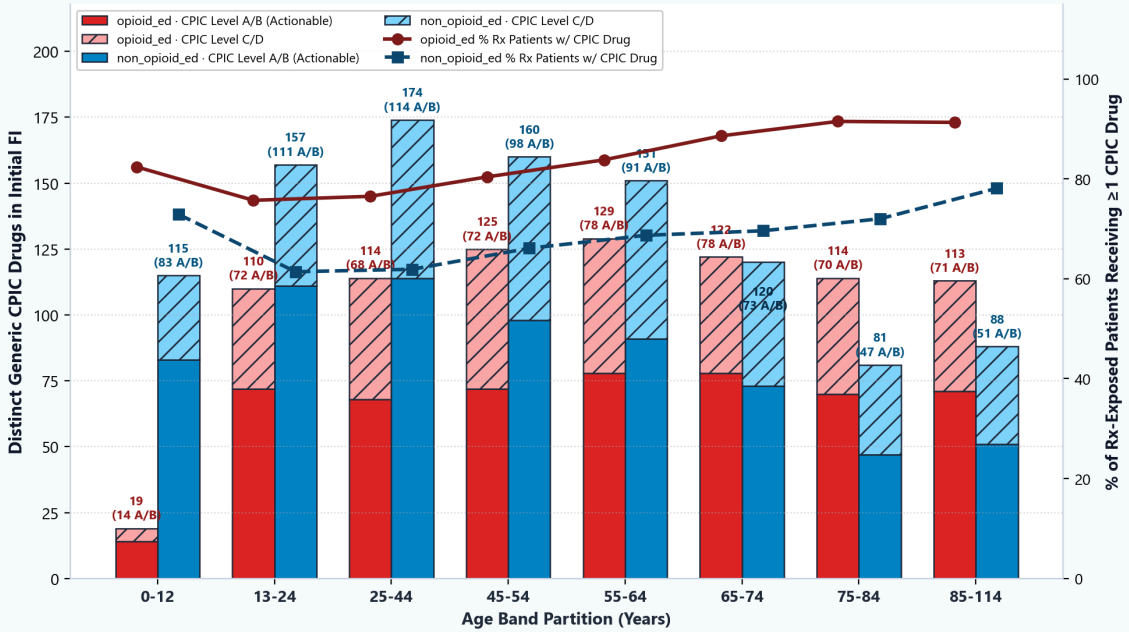
MODULE I | STEP 5

INITIAL EXPLORATORY ANALYSIS (3 OF 3) · CPIC PHARMACOGENOMIC DRUGS IN INITIAL FEATURE IMPORTANCE

Number of CPIC-Actionable Drugs Surfaced in Initial Feature Importance by Target Cohort & Age Band

opioid_ed: 187 Distinct CPIC Drugs (333 Features)
non_opioid_ed: 224 Distinct CPIC Drugs (439 Features)

A. Distinct CPIC Drugs in Initial Feature Importance by Target Cohort & Age Band



B. Dominant CPIC Gene–Drug Pairs in Initial Feature Importance by Target

Pharmacogene	CPIC Lvl	opioid_ed Top FI Drugs (187 CPIC Drugs | 76–92% Rx Pts)	non_opioid_ed Top FI Drugs (224 CPIC Drugs | 61–78% Rx Pts)
CYP2D6 (Analgesics, SSRIs, β-Blockers)	Level A / B	• Oxycodone (#17), Hydrocodone (#14) • Tramadol (#39), Codeine, Ondansetron • Venlafaxine, Risperidone, Aripiprazole	• Tramadol (#3), Hydrocodone (#6) • Metoprolol (#8), Carvedilol, Duloxetine • Ondansetron, Amitriptyline, Mirtazapine
CYP2C19 (PPIs, SSRIs, Antiplatelets)	Level A	• Omeprazole (8/8 bands), Pantoprazole • Sertraline, Citalopram, Escitalopram • Diazepam (8/8 bands), Amitriptyline	• Omeprazole (#9), Pantoprazole (#12) • Escitalopram (#21), Sertraline • Citalopram, Clopidogrel
CYP2C9 & VKORC1 (NSAIDs, Sulfonyl-ureas, Warfarin)	Level A / B	• Ibuprofen (8/8 bands), Meloxicam • Celecoxib, Naproxen, Diclofenac • Phenytoin (CYP2C9 / HLA-B)	• Losartan (#5), Diclofenac, Ibuprofen • Meloxicam, Celecoxib, Glimepiride • Warfarin (CYP2C9 / VKORC1)
SLCO1B1 & ABCG2 (Statins / Hepatic Transporters)	Level A	• Atorvastatin, Simvastatin • Pravastatin, Rosuvastatin • Co-prescribed in ages 45–85+	• Pravastatin (#12), Atorvastatin (#17) • Simvastatin (#18), Rosuvastatin • High geriatric polypharmacy importance
OPRM1 / COMT / UGT2B15 / G6PD	Level B / C	• Buprenorphine/Naloxone (#5) • Fentanyl (#24), Lorazepam (8/8 bands) • Sulfamethoxazole/TMP (8/8 bands)	• Lorazepam (#14), Ofloxacin (#5) • Ciprofloxacin, Sulfamethoxazole/TMP • Nitrofurantoin (G6PD)

EMPIRICAL BRIDGE TO STEP 5 PGx FEATURE ENGINEERING (pgx_num_drugs & pgx_num_cpic_drugs):

1. High CPIC Density in Top Predictors: 24%–35% of all drug features in initial Feature Importance are CPIC-actionable medications (187 in opioid_ed; 224 in non_opioid_ed).
2. Calibrated Scientific Framing: Claims capture CPIC-actionable medication exposure burden (identifying patients whose regimen warrants PGx review), NOT patient genotypes.

pgx_num_drugs & pgx_num_cpic_drugs
Rank #1–#3 Across Final Models

Figure 1B (Part 3): Number of distinct CPIC-actionable drugs in initial Feature Importance results by target cohort (`opioid_ed` vs. `non_opioid_ed`) and age band.

The Frontline Clinical Bottleneck

MODULE I | INFOGRAPHIC PART 4 OF 4 · POINT-OF-CARE TRANSLATION GAP

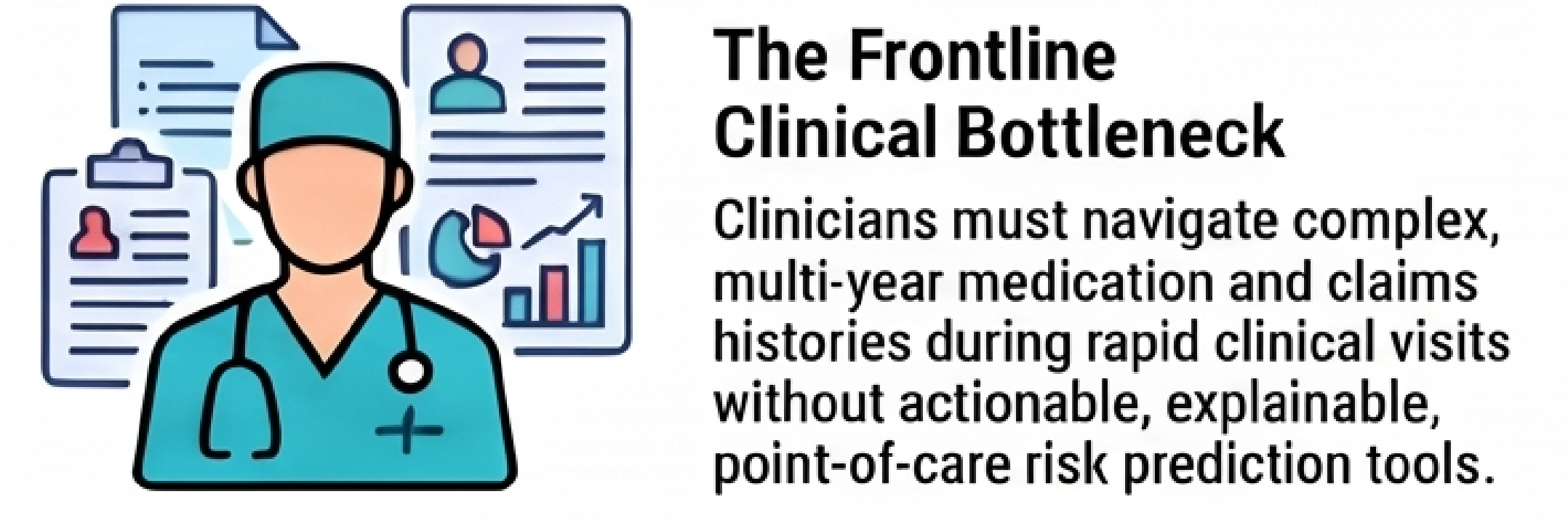


Figure 1A (Part 4): The Frontline & Predictive Clinical Bottleneck across longitudinal medication and administrative claims histories.

AIMS

MODULE I | AIM 1 · COHORT 1 (`opioid_ed`) | AIM 2 · COHORT 2 (`ed_non_opioid`) | AIM 3 · DECISION SUPPORT

Aim 1 · Cohort 1 Opioid ED Prediction & Attribution (Ch. 3)

Develop and temporally validate `CatBoost`, `XGBoost`, and `XGBoost-RF` models with $\mathrm{SHAP} \cap \mathrm{FFA}$ consensus attribution for **Cohort 1** (`opioid_ed` , 12–mo lookback, ages 13–64), evaluating pre-index leakage prevention, discrimination (`PR-AUC` , `AUROC`), and calibration on an independent **2019 temporal holdout**.

Aim 2 · Cohort 2 Polypharmacy & DDI Attribution (Ch. 4)

Develop and temporally validate `CatBoost`, `XGBoost`, and `XGBoost-RF` models with $\mathrm{SHAP} \cap \mathrm{FFA}$ Boolean Horn rules for **Cohort 2** (`ed_non_opioid` , 21–day lookback, ages 65–114) to identify reproducible risk predictors, candidate multi-drug interactions (`IE`), and consensus risk attenuation priorities (`IR`) on the **2019 holdout**.

Aim 3 · Usable Decision-Support Tool (Ch. 5)

Translate validated Aim 1 and Aim 2 risk estimates and interpretable Boolean rules into a usable prototype decision-support tool for pharmacist-led Medication Therapy Management (MTM) review prioritization—evaluating explanation fidelity, clinical interpretability, and workflow usability rather than claiming clinical effectiveness.

Cohort 1: Opioid-Related ED Events (`opioid_ed` , Ages 13–64)

- **Pathophysiology:** Progressive μ-opioid receptor tolerance, escalating MME titration, and synergistic CNS depressants (benzodiazepines, gabapentinoids) blunting respiratory drive.
- **Lookback Horizon: 12–Month Pre–Index Window** capturing chronic prescription escalation and behavioral comorbidities.
- **Claims Anchor (t_0):** Direct diagnostic billing for `F11.20` (opioid dependence) and `T40.0X` – `T40.4X` (opioid poisonings) scanned across **all 10 claim diagnostic positions**.

Cohort 2: Geriatric Polypharmacy & "Silent ADEs" (`ed_non_opioid` , Ages 65–114)

- **Why Pairwise CDS Fails:** Pairwise DDI ($A + B$) checkers trigger **90%–95% alert fatigue** while missing higher-order ($A + B + C + D$) synergies and age-related renal/hepatic decline.
- **The "Silent ADE" Coding Reality:** Fewer than **5%** of drug-induced ED visits are billed with explicit poisoning codes (`T36` – `T50`). Hospitals bill the **downstream clinical syndrome**:
  - **Falls & Syncope:** `R29.6` (falls), `S72` (hip fracture), `R55` (syncope) ← *Z-drugs, benzos, α-blockers*.
  - **Delirium, Bleeds & AKI:** `F05` (delirium), `K92.2` (GI bleed), `N17` (Triple-Whammy AKI).
- **Claims Anchor (t_0):** Milliman HCG emergency lines (`P51b` , `O11` , `P33`) anchored to a **21–Day Proximal Rx Window** (`0-day` guard), with **0 opioid codes** in study year.

Methodological Demarcation: Modeling Cohort 1 (12–month chronic opioid trajectory) and Cohort 2 (21–day proximal geriatric polypharmacy) separately prevents conflating distinct pathophysiologies, age distributions, and coding mechanics.

Governance and Frameworks

MODULE II | PHASE 1 · CRISP-DM, TRIPOD+AI & CLINICAL OODA LOOP

1. CRISP-DM (Systems Lifecycle)

- 6 iterative engineering phases governing the 10-step APCD pipeline.
- Ingestion → Cohort → Whitelist → Model → Eval → Microservice.

2. TRIPOD+AI (Biomedical Validation)

- Transparent reporting; Type 2b / Type 3 temporal validation.
- 2016–2018 train, untouched 2019 calendar holdout (zero post-hoc recalibration).

3. Clinical OODA Loop (Translation)

- **Observe / Orient:** GBT models & formal Boolean Horn rules.
- **Decide / Act:** Decision Support Tool

Appendix A6: Master Acronym & Framework Glossary (CRISP-DM, TRIPOD+AI, OODA) →

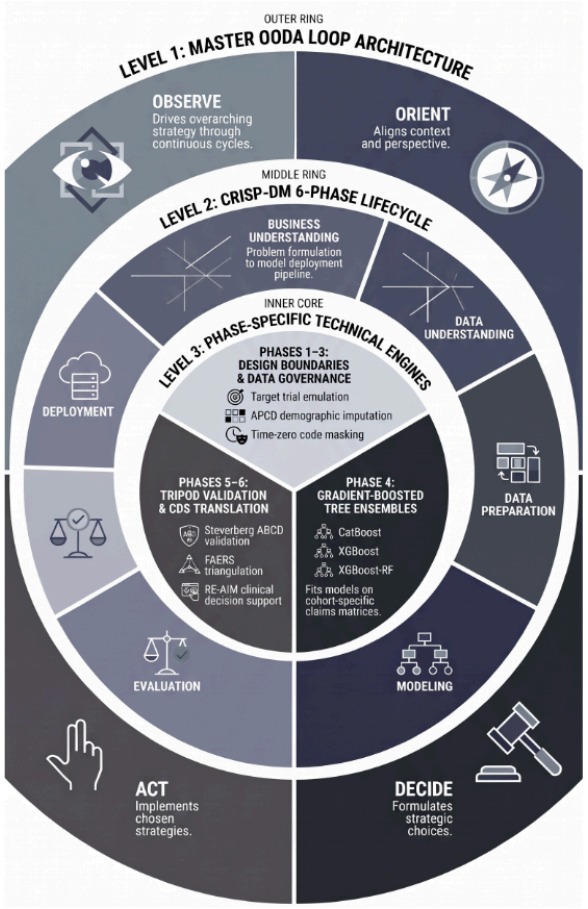


Figure 2: Three-Tier Governance Architecture: Master OODA Loop enclosing 6-phase CRISP-DM and phase-specific technical validation engines.

Why Virginia APCD Over EHR

MODULE II | PHASE 1 · STATEWIDE GENERALIZABILITY & DATA ARCHITECTURE

Virginia APCD (Selected)	**Single-System EHR** (Rejected)
Statewide Capture: Follows patients across health systems (eliminating out-of-network ED blindness).	**Fragmentation:** High loss-to-follow-up when crossing systems. Misclassifies out-of-network overdoses as "healthy non-events."
Medication Reality: Captures adjudicated pharmacy fills, proving physical drug possession.	**Intent vs Exposure:** Captures written prescriptions, missing ~30% primary non-adherence drop-offs.
Linkage: Single honest broker (VHI) guarantees zero probabilistic linkage attrition.	

The Methodological Compensation

We acknowledge the lack of molecular genotypes and real-time vitals in claims by computing CPIC PGx-burden proxies (*CYP2D6/CYP2C19* substrate counts) directly from longitudinal co-prescriptions.

Figure 3: Data Source Architecture: Statewide capture and adjudicated dispenses in Virginia APCD overcome single-system EHR fragmentation and prescription abandonment.

Why Virginia APCD Is Selected

- **Statewide Capture:** Follows patients across health systems, eliminating out-of-network ED blindness.
- **Medication Reality:** Captures paid/adjudicated dispenses, proving physical drug possession.
- **Linkage Integrity:** Single trusted data steward (VHI) guarantees 0 probabilistic linkage attrition.

The Methodological Compensation

- **EHR Genotype Gap:** APCD lacks molecular genotypes and real-time bedside vitals.
- **CPIC Guideline Mapping & Drug Counts:** We compensate by mapping dispensed APCD drugs to whether they carry a CPIC guideline (`drug_cpic_mapping_global.csv`) and counting patient-level total vs. CPIC drugs (`pgx_num_drugs` , `pgx_num_cpic_drugs`) and how many patients carry CPIC drugs.

Medical & Pharmacy Claims Schema

MODULE III | STEP 2

Medical & Pharmacy Relational Join (`mi_person_key`)

- **Medical (`25` cols):** 10 ICD-10 Dx (`primary..ten_icd_diagnosis_code`), 9 ICD-Proc + 3 CPT (`procedure_code` , `cpt_mod_1/2`), HCG (`hcg_line` , `hcg_detail`), imputed demographics.
- **Pharmacy:** Paid dispenses (`incurred_date → event_date`), generic RxNorm `drug_name` , `therapeutic_class_1` .
- **Unified Timeline (`unified_event_fact_table`):** Linked on `(mi_person_key, event_year, age_band)` with demographic backfill & global `event_sequence` ordering.

Key Pre-Index Feature Families ($t < t_0$)

- **Clinical & Rx Indicators:** `item_icd_*` (10 Dx positions), `item_cpt_*` , `item_drug_*` (active generic fills & concurrency).
- **CPIC Drug Mapping & Utilization (`5_pgx_analysis`):** Maps APCD drugs to CPIC guideline status to count patient drugs (`pgx_num_drugs` , `pgx_num_cpic_drugs`) & patients with CPIC drugs, plus `n_events` density strata (`P25/P50/P95`).

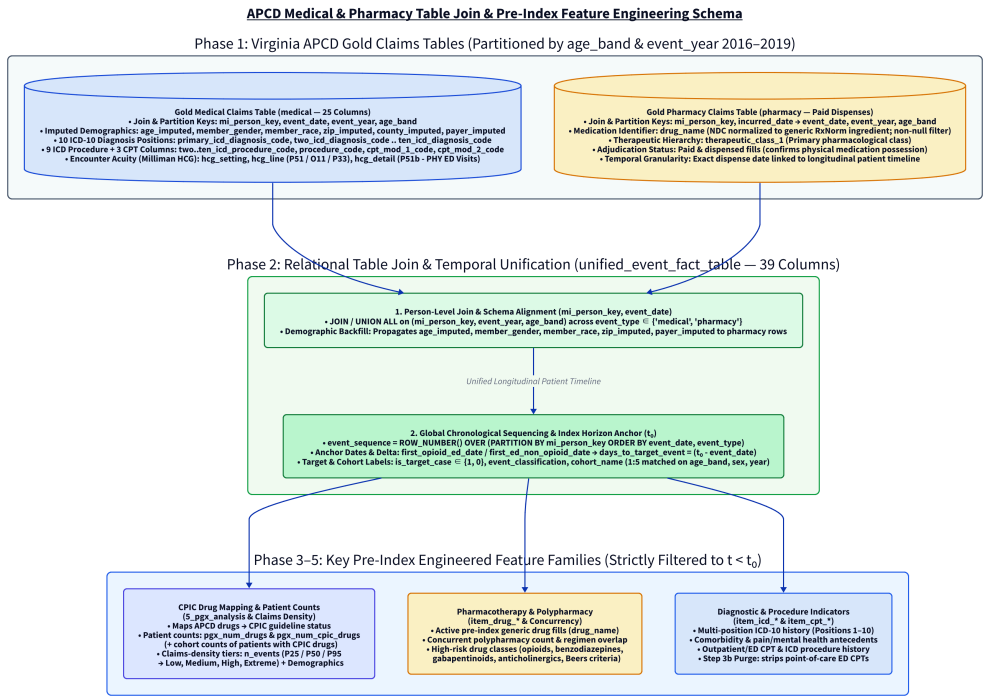


Figure 4: Relational unification of APCD Gold `medical` and `pharmacy` claims into the 39-column `unified_event_fact_table` and pre-index feature matrix.

Schema Integration: Deterministic linkage on `mi_person_key` and global chronological sequencing (`event_sequence`) align diagnoses, procedures, and paid drug fills prior to t_0 .

Cohort 1 Pipeline: Opioid-Related ED (`opioid_ed`)

MODULE III | STEP 2

Clinical Anchor (t_0)

- **ICD-10** `F11.20` (Opioid dependence).
- **Acute Poisonings:** `T40.0X` – `T40.4X`.
- Scanned across all 10 claim positions.

Step 3b Leakage Purge

- Partitions candidate events around t_0 ($t < t_0$ vs. $t \geq t_0$).
- Purges **post-index rescue & MAT induction** (buprenorphine, naloxone) & toxicology screens.
- Locks symmetric pre-index antecedent whitelist.

1:5 Matching & DuckDB QA

- Controls matched on 8 age bands, sex, year.
- Asserts single index date per patient-year.

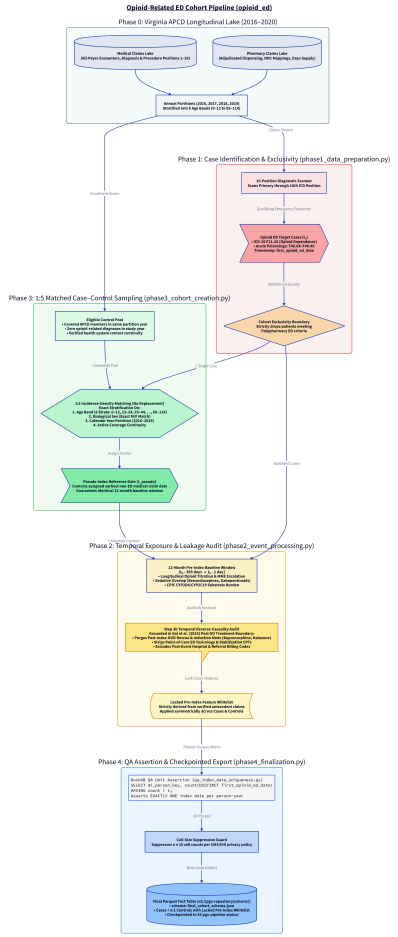


Figure 4A: Opioid-Related ED Cohort Derivation Pipeline (APCD 2016–2020) with 12-month pre-index lookback and locked antecedent whitelist.

Cohort 2 Pipeline: Polypharmacy ADEs (`ed_non_opioid`)

MODULE III | STEP 2

Clinical Anchor (t_0)

- **Milliman HCG Codes:** `P51b`, `011`, `P33`.
- **Absolute Opioid Cleanse:** Zero opioid ICD codes across all 10 positions in study year.

Sequential ADE Funnel

1. **Utilization Cap:** < 7 ED visits/yr.
2. **21-Day Proximity:** Active Rx 1–21d pre-ED (FDA FAERS 90.5%).
3. **0-Day Discharge Guard:** Purges Day 0 fills.

Feature Representation

- 21–30 day acute concurrent exposure.
- Beers criteria & multi-drug regimens.

Cohort Validation FIGURE 1B ↗

- **FAERS vs APCD ADE Events:** >90% kinetic onset convergence at 21 days.
- *Click card to pop out Figure 1B.*

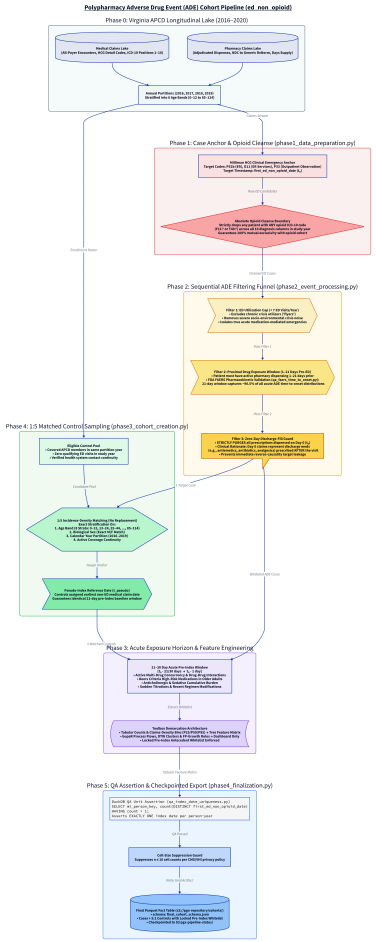


Figure 4B: Polypharmacy Adverse Drug Event Cohort Derivation Pipeline with sequential ADE filtering funnel and 0-day discharge guard.

Gradient-Boosted Trees & Model Selection

MODULE IV STEP 6

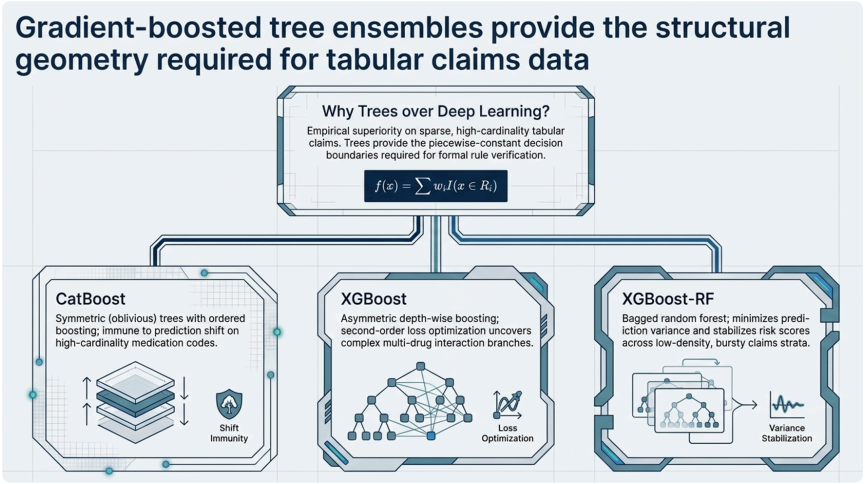


Figure 5: Tri-Model Structural Geometry: Piecewise-constant decision boundaries $f(x) = \sum w_i \mathbb{I}(x \in R_i)$ enabling cross-architecture SHAP consensus and exact CNF rule verification.

CatBoost (Symmetric)

- Oblivious decision trees with ordered boosting.
- Shift immunity on high-cardinality NDC and ICD-10 diagnostic codes.

XGBoost (Asymmetric)

- Depth-wise gradient-boosted decision trees.
- Second-order Hessian loss uncovers complex multi-drug interaction branches.

XGBoost-RF (Bagged)

- Bagged random forest mode with parallel trees.
- Minimizes prediction variance and stabilizes risk across low-density strata.

Step 6 Selection Protocol: Within each cohort, age band, and 4-tier density bin, 25-split MC-CV ranks CatBoost, XGBoost, XGBoost-RF, and Ensemble by **mean PR-AUC** (primary) and **Recall at 0.5** (tiebreaker) to determine the top-performing model for risk scoring and select the winning XGBoost variant plus CatBoost for Step 7 consensus attribution.

Consensus Risk Attribution

MODULE IV | CONSENSUS RISK ARCHITECTURE

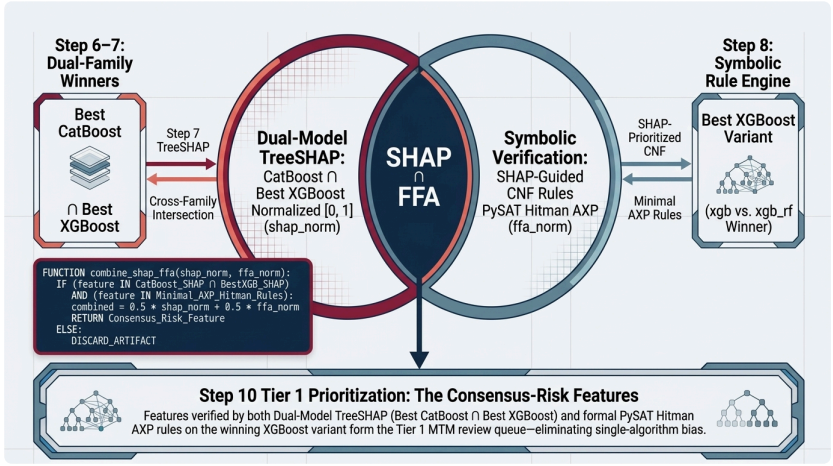


Figure 6A: Aim 2 Consensus-Risk Architecture: Dual-Model TreeSHAP Intersection (`CatBoost` ∩ `Best XGBoost`) guiding Symbolic CNF Rule Extraction on `best_xgb_variant` and Consensus Synthesis.

Dual SHAP Gate & Symbolic CNF

- **Dual-Model `TreeSHAP` Gate:** Computes exact `TreeSHAP` on **best CatBoost** and **best XGBoost variant** (`xgb` vs. `xgb_rf`), enforcing strict feature intersection (`CatBoost` ∩ `Best XGBoost`) and normalized mean score.
- **CNF Compilation:** Explodes root-to-leaf paths of `best_xgb_variant` into Conjunctive Normal Form (CNF) clauses (`_explode_tree_to_dataframe`).
- View Symbolic CNF & PySAT AXP Proof ↗

SHAP-Guided FFA & Synthesis

- **SHAP-Prioritized Rule Extraction:** Rather than extracting all rules from all trees across all models, filters matched CNF rules from `best_xgb_variant` by instance $|\mathrm{SHAP}|$ attribution and active feature coverage → PySAT `Hitman` minimal Abductive Explanations (AXPs; $\mathrm{support} \geq 0.05, \mathrm{confidence} \geq 0.70$).
- **Combined Synthesis:** Ranks drivers via $0.5 \cdot \mathrm{SHAP}_{\mathrm{norm}} + 0.5 \cdot \mathrm{FFA}_{\mathrm{norm}}$ and top-K intersection (`shap_top` ∩ `ffa_top`).

Target Leakage

MODULE IV STEP 3B

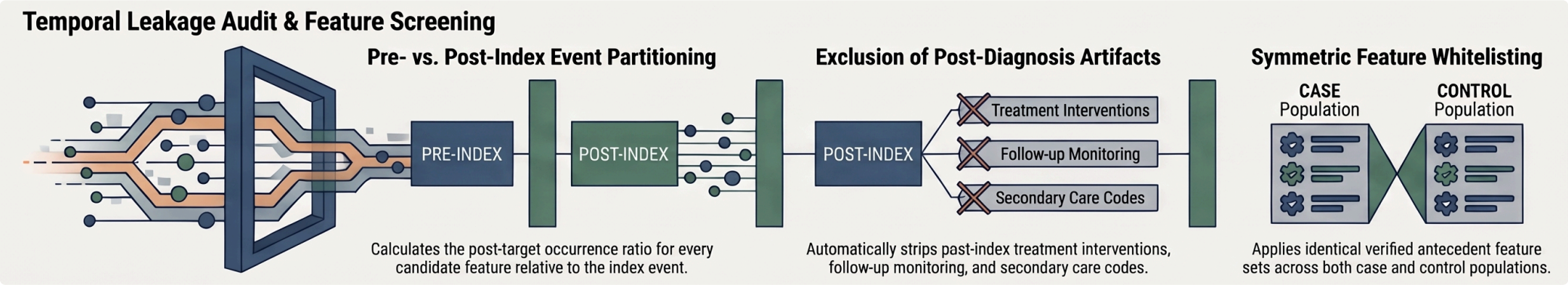


Figure 7: Step 3b Temporal Event-Partitioning Protocol: Pre- vs. post-index ($t < t_0$ vs. $t \geq t_0$) event timing audit, removal of post-event clinical interventions, and symmetric antecedent whitelist locking.

Clinical Rationale (Ochalek, Gal et al., 2023)

- **Post-Overdose Clinical Workflow:** As established in Dr. Gal's VCU ED research (*Ochalek, Gal et al., 2023, DADR*), rescue agents and MAT induction (e.g., buprenorphine, naloxone) are initiated *as a clinical consequence following* an acute overdose.
- **The Reverse-Causality Trap (30% of Literature):** Unmonitored gain rankings (`Step 3a`) capture post-event stabilization and confirmatory toxicology orders occurring at $t \geq t_0$.

Step 3b Symmetric Antecedent Whitelist

- **Temporal Event Partitioning (`bupaR` Audit):** Partitions every candidate feature's timestamps relative to t_0 ($t < t_0$ pre-index vs. $t \geq t_0$ post-index).
- **Automated Post-Target Purge:** Strips candidate codes with $\tau_{\text{post}} \geq 0.80$ (`post_target_ratio ⩾ 0.80` ; post-event MAT induction, ED tox screens, follow-up billing).
- **Immutable Whitelist (`safe_feature_filter.json`):** Locks verified pre-index antecedents symmetrically across cases & controls *(empirical catalogs reserved for Nov 23 Defense)*.
- View Step 3b Event-Partitioning Protocol ↗

Actionable Intelligence and Workflow Integration

MODULE V STEP 10

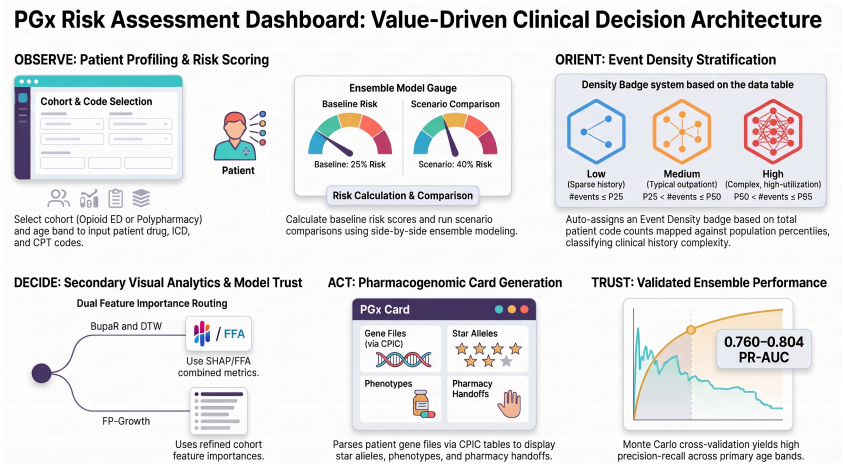


Figure 8: Aim 3 Decision-Support Prototype: Value-Driven Clinical Decision Architecture, OODA Information Delivery, and Event Density Stratification.

Translating Validated Findings for Clinical Review

- **Actionable Triage Prioritization:** Translates Aim 1 risk estimates and Aim 2 consensus features into an interpretable Tier 1–3 medication-review priority list.
- **Explanation Fidelity & Usability:** Evaluates whether concise `IF-THEN` rules accurately reflect model predictions and support rapid pharmacist review.
- **OODA Delivery Framework:** Observe – orient – decide – act framework for delivery of information.
- **Lean Value Stream:** Framework for ranking and layout of information (high to low, left to right by value).

Cognitive Ergonomics vs. 95% Alert Fatigue

- **Overcoming Commercial Pop-ups:** Asynchronous review prioritization replaces **90%–95% overridden** synchronous pop-up interrupts.
- **Sound Symbolic Horn Rule:**

$$\text{IF } [\text{Opioid} \geq 1] \wedge [\text{Benzodiazepine} \geq 1] \wedge [\text{Refill} \leq 14\text{d}] \implies \text{Tier 1 Alert}$$

- **Interactive "What-If" Scenario Exploration:** Displays model-predicted risk shifts when regimen features are toggled while preserving 100% clinician authority.

Dissertation Portfolio & Status

MODULE V | PHASE 6 · DISSERTATION PORTFOLIO & STATUS

Core Empirical Research (Aims 1–3 · Published Papers)

- **Chapter 3 (Specific Aim 1):**
  Temporal Drivers of Opioid ED Encounters in Virginia's APCD
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, 2026; DOI: 10.1111/cts.70690)
  Contribution: Step 3b leakage audit, 12-mo chronic opioid model (`opioid_ed`), 2019 temporal holdout.
- **Chapter 4 (Specific Aim 2):**
  SHAP-Informed Formal Feature Attribution for Polypharmacy ADEs
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, Sep 2026; DOI: 10.1111/cts.70718)
  Contribution: Dual-model `TreeSHAP ∩ FFA` consensus, 21-day acute ADE cohort (`ed_non_opioid`), interaction effects (IE).
- **Chapter 5 (Specific Aim 3):**
  Serverless Clinical Decision Support for Pharmacist Medication Review
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, Oct 2026; DOI: 10.1111/cts.70697)
  Contribution: Zero-PHI cloud prototype on AWS Lambda (<$5/mo), Tier 1–3 triage, overcoming 95% alert fatigue.

Foundational, Methods & Synthesis (Dissertation Status)

- **Chapter 1 (Clinical Foundation & Literature Appraisal):**
  Explainable AI & PGx in High-Risk Prescribing: A Systematic Review
  Status: INVITED REVIEW · *J Pers Med* (MDPI)
  Contribution: Systematic review of 94 ML studies; establishes "The Empty Intersection" (0/73 PROBAST models).
- **Chapter 2 (Scalable Data Architecture & Governance):**
  A Scalable Big-Data Claims Processing Architecture and Governance Lifecycle
  Status: WITH EDITOR FOR REVIEW · *J Big Data* (Springer)
  Contribution: Three-tier governance (CRISP-DM, TRIPOD+AI, Clinical OODA Loop) and DuckDB EC2 claims pipeline (15.1× speedup).
- **Chapter 6 (Translational Synthesis & Dissertation Build):**
  Translational Synthesis, Methodological Defenses, and Clinical Horizons
  Status: COMPLETE DRAFT FOR COMMITTEE REVIEW
  Contribution: Comprehensive synthesis of Chapters 1–5, formal methodological defenses, and prospective clinical validation roadmaps.

Dissertation Status: 3 Published Papers (*Clinical and Translational Science*) • **1 Under Review** (*Journal of Big Data*) • **1 Invited Review** (*Journal of Personalized Medicine*) • **Full Manuscript Drafted** across all 6 chapters for Qualifying Exam and Final Defense.

Study Boundaries

MODULE V | PHASE 6 · 8 INTERLOCKING EPIDEMIOLOGICAL & GOVERNANCE SAFEGUARDS

EPIDEMIOLOGICAL COHORT DESIGN (4 SAFEGUARDS)

1. Absolute Opioid Cleanse

Zero opioid ICD-10 codes across all 10 diagnostic positions in the study year isolates pure geriatric ADEs in `ed_non_opioid`.

2. High-Utilization Cap (<7 ED/yr)

Excludes chronic utilizers with ≥ 7 annual ED visits to prevent high-frequency outlier distortion of feature representations.

3. 21-Day Lookback Window

Active Rx $[t_0 - 21, t_0 - 1]$ empirically validated by convergent FAERS/APCD kinetics (>90% of acute ADEs occur within 21 days).

4. Zero-Day Discharge Guard ($t_{\text{claim}} < t_0$)

Strict timing guard purges day-zero fills (`qa_index_date_uniqueness.py`) and $\tau_{\text{post}} \geq 0.80$ confounders (348 purged).

METHODOLOGICAL & CLINICAL GOVERNANCE (4 SAFEGUARDS)

5. Aim 3: Decision Support, Not an Outcomes Trial

Translates validated model findings into an interpretable prototype tool evaluating explanation fidelity; prospective clinical trials are out of scope.

6. Consensus Risk Attribution, Not Causation

Observational claims and `SHAP ∩ FFA` Consensus Attribution quantify reproducible consensus risk associations rather than causal treatment effects.

7. Independent 2019 Temporal Holdout Validation

Evaluates out-of-time model transportability (`2016-2018` train → `2019` calendar holdout) under secular prescribing shifts with zero recalibration.

8. Claims CPIC Exposure, Not Germline Genotypes

Administrative claims quantify CPIC-actionable medication exposure burden (`pgx_num_drugs`, `pgx_num_cpic_drugs`) rather than inferred patient genotypes.

Methodological Integrity: Eight safeguards unite clinical governance with epidemiological rigor. Crucially, **FDA FAERS kinetic convergence** (>90% ADEs within 21 days) provides external validation for our `ed_non_opioid` cohort logic, while strict controls eliminate target leakage.

Questions & Discussion

Technical Appendix Defenses (A1–A6) →

COMMITTEE Q&A BACKUP BANK

Supplemental Technical Evidence

Mathematical derivations, feature audit catalogs, pharmacogenomic scoring matrices, cloud systems architecture, and reproducible notebook pipelines supporting defense inquiries.

🌐 Open Interactive Web Presentation (pgx.jerome-dixon.io/9NOV_presentation) ↗

Appendix A1: Formal Feature Attribution (FFA) — Best-Model CNF & AXP Bounds

SPECIFIC AIM 2 · STEPS 6–8 & 10 | MATHEMATICAL VERIFICATION & PYSAT AXP

Step 7 Dual-Model Gate & Step 8 CNF Compilation

- Step 7 Dual-Model `TreeSHAP` Gate:

$$\mathcal{F}_{\text{SHAP}} = \text{supp}(\text{SHAP}_{\text{CatBoost}}) \cap \text{supp}(\text{SHAP}_{\text{BestXGB}})$$

  Requires cross-architecture agreement before symbolic extraction (since CatBoost CTR hashing precludes direct symbolic path parsing).
- Step 8 CNF Compilation on `best_xgb_variant`:
  - Parses JSON tree dumps of `best_xgb_variant` (`xgb` vs. `xgb_rf` winner by **PR-AUC** then **Recall**).
  - Explodes root-to-leaf paths into tabular intervals (`_explode_tree_to_dataframe`) and compiles Conjunctive Normal Form (CNF) clauses (`rule_clauses`).

SHAP-Guided Minimal Hitting-Set AXP & Complexity

- SHAP-Guided Rule Filter → PySAT `Hitman`:
  - Rather than brute-forcing $> 10,000$ matched tree rules per patient, prioritizes top SHAP-supported matched rules (with stratified rare-population coverage safeguards) into the PySAT `Hitman` minimal hitting-set solver to extract minimal **Abductive Explanations (AXPs / Horn clauses)** with $\text{Supp}(R) \geq 0.05$ and $\text{Conf}(R) \geq 0.70$.
- Step 8 Attribution Responsibility & Step 10 Combined Synthesis:

$$\text{AttrResp}(f) = \frac{\text{RuleFreq}(f)}{\sum_k \text{RuleFreq}(f_k)} \times \text{SHAP}_{\text{consensus}}(f)$$

$$\text{Score}_{\text{combined}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$$

- **Complexity vs. Neural Nets:** Tree CNF extraction is polynomial $\mathcal{O}(T \cdot L)$, whereas ReLU neural network minimal explanation is **co-NP-complete** (Katz et al., 2017).

↩ Return to Consensus Attribution (Aim 2)

Operational Threshold Validation Protocol (`SHAP ⩾ P75` ∩ `Supp ⩾ 0.05, Conf ⩾ 0.70`): Framed as a tunable *precision–volume dial* for clinical review capacity and validated via a 2-part sensitivity protocol: (1) **Two-Gate Mirror Ablation** (`SHAP-only` vs. `FFA-only` vs. `SHAP ∩ FFA` grid sweep testing top-K core signal retention, boundary noise filtering, and review-load compression); and (2) **Utilization-Free Specification Stress Test** (stripping utilization covariates/density strata to compare univariate `SHAP` Jaccard rank stability vs. symbolic `FFA` rule persistence under model perturbation).

Appendix A2: Step 3b Target Leakage — Temporal Event-Partitioning Protocol

SPECIFIC AIM 1 · STEP 3B | TEMPORAL PROCESS AUDIT

Leakage Vector Category	Representative Claims Signals	Temporal Timing	Clinical Mechanism
Post-Event OUD Rescue & MAT Induction	Buprenorphine, Suboxone, Naloxone, Methadone, Extended-Release MAT	$t \geq t_0$ (Post-Index)	Initiated in ED or bridge clinic *as a consequence following* acute overdose
Confirmatory Toxicology & Monitoring	Presumptive & definitive urine drug screens (e.g., CPT 80307, 80305, G0480)	$t \geq t_0$ (Post-Index)	Ordered during acute ED triage or post-discharge MAT compliance monitoring
Secondary Administrative Follow-Up	Long-term drug therapy status, rehabilitation & subsequent encounter billing	$t \geq t_0$ (Post-Index)	Downstream administrative claims generated directly by the index visit

Step 3b Formal Event-Partitioning Rule & Symmetric Whitelist

- **Post-Target Ratio Filter:** $\tau_{\text{post}}(f) = \frac{N(t_{\text{claim}} \geq t_0)}{N(t_{\text{claim}} < t_0) + N(t_{\text{claim}} \geq t_0)} \geq 0.80 \implies \text{Purge from } \texttt{safe_feature_filter.json}$.
- **Symmetric Pre-Index Enforcement:** All retained features satisfy $t_{\text{claim}} < t_0$ on $[t_0 - 365, t_0 - 1]$ (Opioid) or $[t_0 - 21, t_0 - 1]$ (Polypharmacy) across cases and 1:5 matched controls.

Why Unmonitored Gain Rankings Fail (`Step 3a` vs. `Step 3b`)

- **The Literature Failure (30% of PROBAST Models):**
  - Unfiltered feature selection loads top tree splits onto post-overdose MAT induction and ED toxicology screens, inflating apparent discrimination by 5–20 AUROC points.
- **Candidacy vs. Defense Scope Demarcation:**
  - Today (`9 NOV`): Lock and verify the **Step 3b temporal event-partitioning methodology** and antecedent whitelist.
  - Final Defense (`23 NOV`): Present complete empirical feature audit counts and 2019 holdout discrimination deltas.

↩ Return to Target Leakage Defense

Appendix A3: Step 5 CPIC Drug Mapping & Patient-Level Count Features

SPECIFIC AIM 1 · STEP 5 (`5_pgx_analysis`) CLAIMS EXPOSURE VS. GENOTYPE DEFENSE

1. Global APCD Drug → CPIC Guideline Mapping (`build_global_drug_cpic_mapping.py`)

- **Official CPIC Reference Corpus:** Ingests `cpic_gene-drug_pairs.xlsx` / `cpic_drug_list.json` (`573` gene–drug pairs, `300` unique drugs, `121` pharmacogenes across CPIC Levels A–D).
- **Automated Drug Vocabulary Matching:** Extracts unique APCD drug features (`DRUG:` , `item_drug_*`) from Step 3 feature importance outputs and matches them against CPIC drug names ($\geq 95\%$ fuzzy threshold) to build `drug_cpic_mapping_global.csv` (`s3://pgxdatalake/gold/pgx_features/global/`).
- **Binary Guideline Indicator:** Determines whether or not each dispensed APCD generic medication has an official CPIC pharmacogenomic guideline ($d \in \mathcal{C}_{\text{CPIC}}$).

2. DuckDB Patient-Level Drug & CPIC Counts (`create_pgx_features_patient_level.py`)

- **Per-Patient Count Features (`mi_person_key`):** `pgx_num_drugs` $_i = |D_i|$ (total distinct pre-index drugs) and `pgx_num_cpic_drugs` $_i = |\{d \in D_i : d \in \mathcal{C}_{\text{CPIC}}\}|$ (distinct CPIC-guideline drugs).
- **Cohort Exposure Counts:** Aggregates distinct pre-index drugs (D_i) per patient in DuckDB and records cohort-level counts of patients with any drug (`pgx_num_drugs` $_i > 0$) vs. patients carrying CPIC-guideline drugs (`pgx_num_cpic_drugs` $_i > 0$).

3. Claims-Derived Exposure vs. Genotype Imputation Defense

- **Step 5 Supervised Model Pipeline (`5_pgx_analysis`):**
  - Uses **observable CPIC drug counts only** (`pgx_num_drugs` , `pgx_num_cpic_drugs`) merged via `add_pgx_features_to_model_data.py` .
  - **Claims Boundary:** Administrative claims quantify dispensed drug exposure, not molecular genotypes or germline DNA sequences.
- **Strict Exclusion of Race-as-Proxy Imputation:**
  - Administrative race codes are strictly excluded as an unscientific, inaccurate proxy for genetic ancestry or metabolizer status.
  - Preserves methodological integrity: models test whether actionable medication burden predicts ED risk without asserting unmeasured biology.
- **Scope Boundary (Candidacy vs. Final Defense):**
  - Candidate models evaluate observable claims exposure directly; interactive downstream point-of-care patient genotype and diplotype cards are staged for the **November 23 Final Dissertation Defense**.

↩ Return to Claims Join & Schema ↩ Return to Model Selection

Appendix A4: Computational Pipeline & Notebook Architecture

AIM 1 AIM 2 AIM 3 FULL TRACEABILITY

Jupyter Notebook Pipeline Directory (`pgx-analysis`)

Notebook	Steps	Core Functions & Deliverables
`0_config_and_pipeline.ipynb`	Setup	Environment config, S3-to-NVMe sync, DuckDB data dictionary
`1_cohort_workflow.ipynb`	1a, 1b, 2	FAERS & APCD 21-day ADE window calibration, dual cohort extraction, 1:5 incidence-density matching, 0-day guard
`2_feature_importance.ipynb`	3a, 3b, 3c	Feature screening, Step 3b BupaR event-partitioning audit, safe antecedent whitelist
`3_model_train_shap_ffa.ipynb`	4, 5, 6, 7, 8	Density tiers, CPIC drug mapping & counts (`pgx_num_cpic_drugs`), Step 6 best-model selection, TreeSHAP & FFA rules
`3_model_sensitivity.ipynb`	CH4 Supp	Utilization-free sensitivity & claims-density stratification checks
`4_dashboard_visuals.ipynb`	9	BupaR clinical pathways, DTW trajectory clusters, FP-Growth itemsets
`5_build_and_deploy.ipynb`	10	Multi-stage Docker packaging, AWS Lambda deployment, API Gateway

Execution Model, Checkpoints & Reproducibility

- **S3 Checkpoint Idempotency:**
  - Each stage syncs inputs from `s3://pgx-analysis-data/` to local NVMe.
  - SHA-256 state markers prevent redundant compute on completed steps.
- **Dual Compute Environments:**
  - **Local Workstation:** Windows 11, DuckDB 1.4+, Python 3.11, R 4.3 (bupaR).
  - **Batch Compute:** Amazon Linux 2023 EC2 (`mushin` account `535362115856`).
- **Source Control & Public Artifacts:**
  - Monorepo: `github.com/Jerome3590/pgx-analysis`

↩ Return to Dissertation Portfolio

Appendix A5: Specific Aims, Methodological Estimators & Evaluation Metrics

SPECIFIC AIM 1 | SPECIFIC AIM 2 | SPECIFIC AIM 3 | PRE-RESULTS METHODOLOGICAL & MATHEMATICAL SPECIFICATION

Specific Aim & Pipeline Stage	Pre-Specified Design & Transformation	Mathematical Estimator & Governance Gate
Aim 1 · Steps 1–2: Matching & `0-Day` Guard	`1:5` incidence-density control sampling on `(age_band, sex, year)`; lookback $[t_0 - W,\ t_0 - 1]$	$t_{\text{claim}} < t_0 \quad (W = 365\text{d}$ `opioid_ed`$, W = 21\text{d}$ `ed_non_opioid`$)$ (purges $t = t_0$ discharge fills)
Aim 1 · Step 3b: Leakage Ratio $\tau_{\text{post}}(c)$	Event-level `bupaR` pre/post index (t_0) timing partition across candidate claims	$\tau_{\text{post}}(c) = \frac{N_{t \geq t_0}(c)}{N_{t<t_0}(c) + N_{t \geq t_0}(c)} < 0.80$ (locks symmetric `safe_feature_filter.json`)
Aim 1 · Step 4: Density Strata & Burstiness B	Training (`2016-2018`) pre-index claim quantiles ($P_{25},\ P_{50},\ P_{95}$); inter-claim gaps Δt	$b(n_{\text{ev}}) \in \{0,\ 1,\ 2,\ 3\}$ (`low` - `extreme`); $B = \frac{\sigma_{\Delta t} - \mu_{\Delta t}}{\sigma_{\Delta t} + \mu_{\Delta t}} \in [-1,\ 1]$
Aim 1 · Step 5: CPIC Drug Mapping & Counts	Global APCD drug → CPIC guideline mapping (`drug_cpic_mapping_global.csv`) & patient CPIC drug counts	`pgx_num_drugs`$_k = \lvert D_k \rvert$; `pgx_num_cpic_drugs`$_k = \lvert \{d \in D_k : d \in \mathcal{C}_{\text{CPIC}}\} \rvert$
Aim 1 · Step 6: MCCV Ensemble & Holdout	25-split stratified MCCV (`CatBoost`, `XGB`, `XGB-RF`) + Platt scaling on untouched `2019` holdout	$\hat{p}_{\text{ens}}(\mathbf{x}) = \sum_{m=1}^{3} w_m \hat{p}_m(\mathbf{x}), \quad w_m = \frac{\text{PR-AUC}_m}{\sum_j \text{PR-AUC}_j}$; $\text{Lift} = \frac{\text{AUPRC}}{\pi}, \quad \text{BS} = \frac{1}{N}\sum (\hat{p}_k - y_k)^2, \quad \text{ICI}$

Specific Aim & Attribution Stage	Pre-Specified Design & Demarcation	Mathematical Estimator & Governance Gate
Aim 2 · Steps 7–8: Consensus Gate $\mathcal{C}$ ★	Dual-model `TreeSHAP` (`Cat ∩ Best XGB`) + PySAT `Hitman(m22)` minimal hitting-set `AXP`	$\overline{\lvert \phi_j \rvert} \geq Q_{75} \ \cap\ \text{Supp}_{\text{AXP}}(f_j) \geq 0.05$ $\cap\ \text{Conf}(r) = \frac{N(r \wedge \hat{y}=1)}{N(r)} \geq 0.70$
Aim 2 · Step 8: Intervention Rate (`IR`) ★	In-model consensus risk sensitivity contrast ($f_i \to x_{\text{med}}$, non-causal); **never from FP-Growth**	$\text{IR}_{\Delta\hat{p}}(f_i) = \frac{1}{n}\sum_{k=1}^{n}\left(\hat{p}^{(k)} - \hat{p}^{(k)}_{f_i \to \text{med}}\right)$; $\text{IR}_{\text{AXP}}(f_i) = \frac{1}{\lvert E_1 \rvert}\sum_{k \in E_1} \mathbb{I}\left(\hat{y}_{f_i \to \text{med}} = 0\right)$
Aim 2 · Step 8: Interaction Effect (`IE`) ★	In-model consensus risk contrast Δ_{11} & Class-1 `AXP` co-occurrence lift (non-causal; 1,000 bootstrap CIs)	$\Delta_{11} = \mathbb{E}[\hat{p}_{11}] - \mathbb{E}[\hat{p}_{10}] - \mathbb{E}[\hat{p}_{01}] + \mathbb{E}[\hat{p}_{00}]$; $\text{IE}(f_i, f_j) = \frac{P(f_i \wedge f_j \mid \text{AXP}_1)}{P(f_i \mid \text{AXP}_1)\,P(f_j \mid \text{AXP}_1)} > 1.0$
Aim 3 · Step 9: Visual Analytics Demarcation	Training-cohort `bupaR`, `DTW`, & `FP-Growth`; **strictly excluded from predictor matrix**	$D_{\text{DTW}} = \min_{\pi \in \mathcal{A}_w} \sqrt{\sum \lVert \mathbf{x}_{i,t_1} - \mathbf{x}_{j,t_2} \rVert_2^2}$; $\text{Lift}_{\text{FPG}}(A \to B) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A)\,\text{Supp}(B)}$
Aim 3 · Step 10: OODA Scenario Delta $\Delta\hat{p}$	Stateless consensus risk scenario delta ($\mathbf{x} \to \mathbf{x}'$); Imputation of Normality ($x_{\text{unobs}} = 0$)	$\Delta\hat{p} = \hat{p}_{\text{ens}}(\mathbf{x}') - \hat{p}_{\text{ens}}(\mathbf{x})$; $\text{Score}_{\text{comb}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$

Candidacy (`9 NOV`) vs. Final Defense (`23 NOV`) Scope Demarcation: Today's examination locks the pre-specified study designs, mathematical estimators, and strict predictive-vs.-visual boundary across Specific Aims 1–3. Complete empirical 2019 holdout outcomes, feature-level `IR` / `IE` rankings, and post-execution sensitivity findings are staged for the November 23 Final Dissertation Defense.

↩ Return to Consensus Attribution (Aim 2) | ↩ Return to Methodological Fences | Appendix A6: Master Acronym Reference →

Appendix A6: Master Acronym & Technical Nomenclature Reference

GOVERNANCE: CRISP-DM · TRIPOD+AI · OODA | APCD · HCG (MILLIMAN) · CPIC PGX | GBDT · TREESHAP · PYSAT FFA/AXP | SERVERLESS CDS & MTM

Acronym	Full Expansion	Operational Role in Dissertation
`CRISP-DM`	Cross-Industry Standard Process for Data Mining	6-phase systems engineering lifecycle governing Steps 1a–10 (Business Understanding → Data Prep → GBDT Modeling → Lambda Deployment).
`TRIPOD+AI`	Transparent Reporting of a Multivariable Prediction Model (+ AI)	27-item biomedical validation standard enforcing Type 2b/3 temporal split (`2016-2018` train → untouched `2019` holdout; zero recalibration).
`OODA` **Loop**	Observe-Orient-Decide-Act Loop	Clinical workflow boundary: GBDT & Horn rules *Observe/Orient*; human pharmacists retain 100% authority to *Decide/Act*.
`PROBAST` / `SQLR`	Prediction Model Risk of Bias Assessment Tool / Systematic Quant. Lit. Review	Chapter 1 PRISMA 2020 review ($n = 94$ ML studies; $n = 73$ PROBAST models) exposing the **"Empty Intersection"** ($n = 0/73$).
`APCD` / `VHI`	All-Payer Claims Database / Virginia Health Information	Statewide longitudinal medical & adjudicated pharmacy claims lake (`2016-2019` , 4.2M patients, 380M claims, deterministic `mi_person_key`).
`HCG` **(Milliman)**	Milliman Health Cost Guidelines (HCG Grouper)	Standardized APCD encounter acuity taxonomy (`hcg_setting` , `hcg_line` , `hcg_detail`) anchoring ED visits: `P51` / `P51b` (PHY ED), `O11` (ER), `P33` (Urgent Care).
`CPIC` / `PGx`	Clinical Pharmacogenetics Implementation Consortium / Pharmacogenomics	Step 5 maps APCD drugs to CPIC guideline status (`drug_cpic_mapping_global.csv`) to quantify patient-level CPIC medication exposure burden (`pgx_num_drugs` , `pgx_num_cpic_drugs`).
`ADE` / `DDI` / `FAERS`	Adverse Drug Event / Drug-Drug Interaction / FDA Adverse Event Reporting System	Geriatric **"Silent ADEs"** (`R29.6` falls, `S72` hip fx, `R55` syncope, `N17` AKI) & FDA FAERS 21-day onset kinetics ($> 90\%$ acute ADEs).
`OUD` / `MME` / `MAT`	Opioid Use Disorder / Morphine Milligram Equivalents / Meds for Addiction Tx	12-month chronic opioid trajectory anchors (`F11.20` , `T40.0X-T40.4X`) vs. post-index rescue drugs purged by Step 3b (buprenorphine, naloxone).

Acronym	Full Expansion	Operational Role in Dissertation
`GBDT` / `XGB-RF`	Gradient-Boosted Decision Trees / XGBoost Random Forest	Piecewise-constant ensembles comparing oblivious trees (`CatBoost`), depth-wise trees (`XGBoost`), and bagged trees (`XGBoost-RF`).
`MCCV` / `TPE`	Monte Carlo Cross-Validation / Tree-structured Parzen Estimator	25-split stratified development CV & Bayesian hyperparameter optimization (`Optuna` , 50 trials) with Platt probability calibration.
`AUROC` / `PR-AUC` / `ICI`	Area Under ROC / Precision-Recall AUC / Integrated Calibration Index	Discrimination, imbalanced minority-class precision-recall lift, Brier Score, ICI calibration, and Decision Curve Analysis (`DCA`).
`XAI` / `TreeSHAP`	Explainable AI / Tree SHapley Additive exPlanations	Exact game-theoretic feature attribution (ϕ_j) intersected across `CatBoost` ∩ `Best XGBoost` ($\overline{\lvert\phi_j\rvert} \geq Q_{75}$) in Step 7.
`FFA` / `CNF` / `PySAT`	Formal Feature Attribution / Conjunctive Normal Form / Python SAT Solver	Step 8 compilation of tree paths into CNF clauses solved via PySAT `Hitman` minimal hitting sets in polynomial $\mathcal{O}(T \cdot L)$ time.
`AXP` **& Horn Rules**	Abductive Explanation & Boolean Horn Logic Clauses	Minimal irreducible `IF-THEN` conditions ($P_1 \wedge P_2 \implies \hat{y} = 1$) guaranteeing Class-1 risk ($\text{Supp} \geq 0.05, \text{Conf} \geq 0.70$).
`IR` & `IE` (Δ_{11})	Intervention Rate & Interaction Effect (2nd-Order Risk Contrast)	In-model consensus risk attenuation drop ($\text{IR}_{\Delta\hat{p}}, \text{IR}_{\text{AXP}}$) and cross-partial synergy lift ($\Delta_{11}, \text{IE} > 1.0$) strictly from trees/AXPs.
`bupaR` / `DTW` / `FPG`	Business Process Analysis in R / Dynamic Time Warping / FP-Growth	Exploratory process flows, trajectory distance (D_{DTW}), and itemset lift (Lift_{FPG}) **strictly excluded from supervised GBDT matrices**.
`CDS` / `MTM` / `PHI`	Clinical Decision Support / Medication Therapy Mgmt / Protected Health Info	Aim 3 stateless, zero-PHI AWS Lambda microservice ($\mu_{\text{warm}} = 6\text{ms}, < \$5/\text{mo}$) driving an asynchronous Tier 1–3 pharmacist review queue.

↩ Return to Governance and Frameworks | ↩ Return to Polypharmacy Cohort Pipeline | ↩ Return to Methodological Fences